Song 2019 NSCLC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

3 tumour samples (P4/P3/P1, ≥100-cell floor after dropping the 3-cell P4_Normal spillover) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→1.65 of 3); a single Leiden lobe carries the whole trajectory.

Gate 2 — module geography → GO (non-branching)

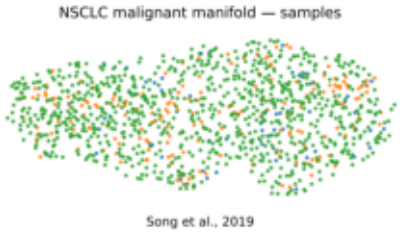
Single dominant (Gast/pSom)→Crest axis: Gast/pSom high over the origin decline along pseudotime (Gast ρ=−0.72, pSom ρ=−0.37); Crest anti-correlated in space (Gast↔Crest r=−0.89), rising to a terminal readout (ρ=+0.74; Crest-high decile mean pt=0.84). Joint pSom&Crest-high depleted (enrichment 0.54) — an inverse gradient, not a co-expression hub.

Gate 3 — cross-repression → NA

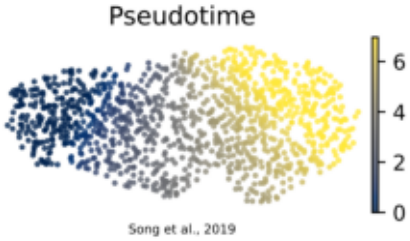
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

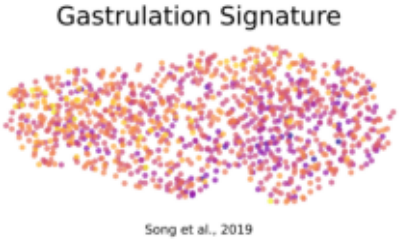
Manifold — samples



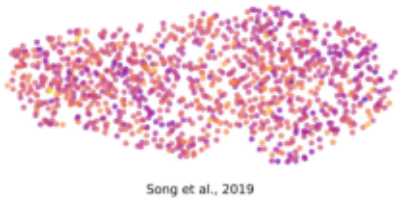
Pseudotime



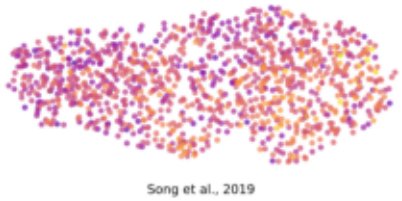
Gastrulation



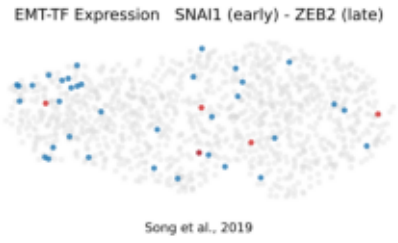
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

NSCLC / lung adenocarcinoma (Song et al., 2019), 10x — 6 samples / 8,772 cells (master scope); malignant only: 1,144 cells across 3 tumour samples (drop the 3-cell P4_Normal spillover); whole malignant manifold (single lobe, no scoping). Non-branching Linear-pSom/Crest.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Song's NSCLC malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient rooted at the Gast/pSom-high pole; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The compartment is 57% cycling but forms one Leiden lobe — the axis is developmental, not proliferative (Gast/Crest cell-cycle-free; Gast↔Crest anti-correlation survives within non-cycling cells, r=−0.87; pseudotime↔cycling ρ=−0.19). EMT-TFs are detection-limited in this sparse 1.1k-cell compartment: only SNAI1 (root, 12.5%) and ZEB2 (terminus, 1.6%) survive the gene filter.

Parameter choices

Min cells/sample	√(62·3)≈13 (62→3 cliff) → 3 tumour samples, 1,144 cells	**Scope**	whole manifold (single lobe, no scoping)
Root node	tip 2 = Gast/pSom-high	**Integration**	Harmony θ=10; n_pcs=20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Geometry**	Linear-pSom/Crest (non-branching)
EMT-TFs	SNAI1 (root, 12.5%) → ZEB2 (terminus, 1.6%); rest below detection		